The Pentagon, Machine-Checked

Abel's five-term relation for the dilogarithm in Lean 4

Carles Marín*

Independent researcher | karlesmarin@gmail.com

ORCID 0009-0007-5637-9688

June 2026

Abstract

We give the first machine-checked proof, in any proof assistant, of *Abel's five-term relation* for the dilogarithm. In its cleanest form, with L the Rogers L-function, the relation reads, for $x, y \in (0,1)$,

$$\mathrm{L}(x) + \mathrm{L}(y) = \mathrm{L}(xy) + \mathrm{L}\left(\tfrac{x(1-y)}{1-xy}\right) + \mathrm{L}\left(\tfrac{y(1-x)}{1-xy}\right).$$

This single identity is the capstone of the weight-2 theory: by theorems of Wojtkowiak and de Jeu, *every* functional equation of the dilogarithm with rational arguments is a consequence of it. The formalization is in Lean 4 over Mathlib, built on a self-contained development of Li_2 (Mathlib carries no dilogarithm). The proof is the classical differentiation argument made fully rigorous: fixing y, the five-term difference has identically zero derivative on $(0,1)$ – an algebraic cancellation of nine logarithmic terms into a five-element basis – so it is constant, and its boundary value as $x \to 0^+$ is computed by squeezing $\log x \cdot \log(1-x) \to 0$. The result is `sorry`-free, depending only on `propext`, `Classical.choice` and `Quot.sound`.

1 The most important functional equation

The dilogarithm $\mathrm{Li}_2(x) = \sum_{n\geq 1} x^n/n^2$ obeys a small zoo of functional equations. Two are elementary – Euler's *reflection* $\mathrm{Li}_2(x) + \mathrm{Li}_2(1-x) = \frac{\pi^2}{6} - \log x \log(1-x)$ and *Landen*'s transformation – and one is not: the *five-term relation*, found by Spence (1809) and Abel (1830) and rediscovered many times since. It is the deepest of the weight-2 identities, and the only one that genuinely needs two variables.

Its importance is not folklore. Wojtkowiak proved that every one-variable functional equation of Li_2 with arguments rational in the variable is a $\mathbb{Z}$-linear combination of instances of the five-term relation; de Jeu extended this to arbitrarily many variables [1, 2]. The same identity is the defining relation of the *Bloch group*, hence governs scissors-congruence invariants and the volumes of hyperbolic 3-manifolds [3]; in the theory of cluster algebras it is the *pentagon*, the relation satisfied by a sequence of five mutations [4]. When one writes the five arguments in a cycle (Figure 1) the relation acquires a five-fold symmetry that is the source of all of this structure.

The cleanest statement uses the *Rogers* L*-function* $\mathrm{L}(x) := \mathrm{Li}_2(x) + \frac{1}{2}\log x \log(1-x)$, in which the spurious logarithmic terms of the raw Li_2 form are absorbed and the relation becomes a pure sum of five L values, with no elementary correction:

*Formalized with AI assistance (Claude, Anthropic); the mathematics and all claims are the author's responsibility. The exact boundary of the contribution is stated in Section 4.

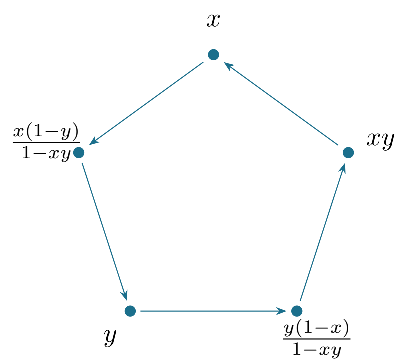

Figure 1: The five arguments of the relation on a cycle. The five-fold symmetry of this pentagon is the combinatorial shadow of the Bloch-group relation and of the cluster-algebra pentagon.

$$\boxed{\mathrm{L}(x) + \mathrm{L}(y) = \mathrm{L}(xy) + \mathrm{L}\Big(\tfrac{x(1-y)}{1-xy}\Big) + \mathrm{L}\Big(\tfrac{y(1-x)}{1-xy}\Big), \qquad x, y \in (0,1).}$$

The state of formalization. Mathlib, the largest formal mathematics library, carries no dilogarithm at all: neither Li_2 nor any of its functional equations. A check of the Lean, Isabelle and Rocq (Coq) libraries in June 2026 turned up no formalization of the five-term relation, or indeed of Li_2 as an analytic object beyond elementary special values. The development reported here is therefore self-contained: it sits on top of a from-scratch Lean construction of Li_2 (its series, derivative $-\log(1-x)/x$, reflection, Landen and duplication), of which the five-term relation is the capstone. With it, the two-variable identities and the polylogarithm *ladders* – evaluations such as the golden-ratio and quartic-base ladders – become reachable as corollaries rather than separate formalization targets.

2 The statement, and the idea of the proof

Theorem 1 (`Dilog.rogersL_fiveterm`)**.** *For all $x, y \in (0,1)$,*

$$\mathrm{L}(x) + \mathrm{L}(y) = \mathrm{L}(xy) + \mathrm{L}\Big(\tfrac{x(1-y)}{1-xy}\Big) + \mathrm{L}\Big(\tfrac{y(1-x)}{1-xy}\Big).$$

The proof is the textbook differentiation argument, carried out with full rigour. Fix $y \in (0,1)$ and set

$$G(t) = \mathrm{L}(t) - \mathrm{L}(ty) - \mathrm{L}\Big(\tfrac{t(1-y)}{1-ty}\Big) - \mathrm{L}\Big(\tfrac{y(1-t)}{1-ty}\Big).$$

The theorem is the statement $G(x) = -\mathrm{L}(y)$. The argument has three movements.

1. The derivative vanishes. For $x, y \in (0,1)$ all three composite arguments lie in $(0,1)$ as well, so each term is differentiable. Writing $\mathrm{L}'(t) = -\frac{1}{2}\big(\frac{\log(1-t)}{t} + \frac{\log t}{1-t}\big)$, the chain rule turns $G'(t)$ into a sum of four such expressions weighted by the derivatives of the composite arguments. Every logarithm that appears – of t, of ty, of the two fractional arguments, and of their complements – expands, by $\log(ab) = \log a + \log b$ and $\log(a/b) = \log a - \log b$, into the five-element basis

$$\{\log t,\ \log y,\ \log(1-t),\ \log(1-y),\ \log(1-ty)\}.$$

In that basis $G'(t)$ is a rational expression whose every coefficient cancels: $G' \equiv 0$ on $(0,1)$. In Lean this is the single delicate step (`fiveterm_hasDerivAt_zero`): the nine logarithmic terms are rewritten into the basis and the rational remainder is cleared and closed by `ring`; the cancellation was checked numerically to thirty digits before it was formalized.

2. Hence G is constant. A function with vanishing derivative on the open, connected interval $(0,1)$ is constant there (Mathlib's `IsOpen.is_const_of_deriv_eq_zero`, the mean-value theorem in disguise). So $G(x) = G(t)$ for every $t \in (0,1)$.

3. The boundary value. It remains to evaluate the constant. As $t \to 0^+$, $ty \to 0$, $\frac{t(1-y)}{1-ty} \to 0$, and $\frac{y(1-t)}{1-ty} \to y$, so

$$G(t) \;\longrightarrow\; \mathrm{L}(0) - \mathrm{L}(0) - \mathrm{L}(0) - \mathrm{L}(y) = -\,\mathrm{L}(y),$$

provided L is right-continuous at 0 with $\mathrm{L}(0) = 0$. This is the one analytic subtlety: $\mathrm{L}(t) = \mathrm{Li}_2(t) + \frac{1}{2}\log t \log(1-t)$, and while Li_2 is continuous, the product $\log t \cdot \log(1-t)$ is an indeterminate $(-\infty)\cdot 0$. It is tamed by a squeeze (`tendsto_rogersL_nhdsGT_zero`): on $(0, \frac{1}{2}]$ one has $-\log(1-t) \le 2t$ (from $\log z \ge 1 - z^{-1}$), so

$$0 \;\le\; \log t \cdot \log(1-t) = (-\log t)(-\log(1-t)) \;\le\; 2t\,(-\log t) = 2\,(-t\log t) \;\longrightarrow\; 0,$$

because $t \log t \to 0$ – Mathlib's `Real.continuous_mul_log`. Combining, G tends to $-\,\mathrm{L}(y)$; being constant and equal to $G(x)$ near 0, it *equals* $-\,\mathrm{L}(y)$. Rearranging $G(x) = -\,\mathrm{L}(y)$ gives Theorem 1. $\square$

The proof in one breath

Fix y. Differentiate the five-term difference in x: the logs cancel into a basis of five and the rest is `ring`. Zero derivative on a connected interval means constant. Slide x to 0: three terms vanish, one survives as $\mathrm{L}(y)$, and the constant is $-\,\mathrm{L}(y)$.

3 The building blocks

The headline theorem rests on three reusable pieces, each `sorry`-free with the standard three axioms.

Lemma 1 (`Dilog.hasDerivAt_rogersL`)**.** *On* $(0,1)$, $\mathrm{L}'(x) = -\frac{1}{2}\Big(\frac{\log(1-x)}{x} + \frac{\log x}{1-x}\Big)$.

This is the same derivative scheme already used in the Euler-reflection and Landen formalizations: combine the derivative $-\log(1-x)/x$ of Li_2 with the product rule for $\frac{1}{2}\log x \log(1-x)$.

Lemma 2 (`Dilog.tendsto_rogersL_nhdsGT_zero`)**.** $\mathrm{L}(t) \to 0$ *as* $t \to 0^+$.

The Li_2 part is continuous on $[-1,1]$; the log-product is squeezed to 0 by $2\,(-t\log t)$ as above. This is the only place the proof leaves pure algebra for genuine analysis.

Lemma 3 (`Dilog.continuousAt_rogersL`)**.** L *is continuous at every* $x \in (0,1)$.

Needed for the surviving boundary term $\mathrm{L}\left(\frac{y(1-t)}{1-ty}\right) \to \mathrm{L}(y)$: Li_2 is continuous at interior points, and $\log x$, $\log(1-x)$ are continuous where their arguments are positive.

4 The boundary of the contribution

Everything here that is mathematics is classical. The five-term relation is due to Spence and Abel; the Rogers L-function and its role in cleaning the identity are from Rogers (1907); the universality statements are Wojtkowiak's and de Jeu's; the differentiation proof is standard. We claim no new identity and no new inequality.

The contributions are three. First, the *formalization itself*: to the author's knowledge (checks of June 2026) this is the first machine-checked proof of the five-term relation in any proof assistant, and it is built on the first machine-checked dilogarithm. Second, the *reusable infrastructure* – the Rogers-L derivative, the right-continuity at 0 via the $-t \log t$ squeeze, and the interior continuity – which is exactly what any further weight-2 development (ladders, the Bloch group, the five-term in Bloch–Wigner form) will need. Third, the *proof-engineering record*: the cancellation of nine logarithms into a five-element basis is the kind of step where a formal check earns its keep, and the boundary limit is a small but instructive piece of elementary asymptotics rarely written out in full.

On methodology. The formalization was carried out with an AI assistant (Claude, Anthropic), with the central cancellation verified numerically to thirty digits before formalization and every named theorem checked by `#print axioms` to depend only on `propext`, `Classical.choice` and `Quot.sound`. Correctness does not rest on the assistant: the Lean kernel checks every proof. The mathematics and the claims remain the author's responsibility.

5 Open roads

1. **Ladders as corollaries.** With the five-term relation in hand, the golden-ratio ladders $\mathrm{L}(1/\varphi) = \pi^2/10$, $\mathrm{L}(1/\varphi^2) = \pi^2/15$ and their quartic-base relatives are linear-algebra consequences rather than separate proofs; making this mechanical in Lean is the natural next step.

2. **The Bloch–Wigner five-term.** The same relation holds for the single-valued Bloch–Wigner function D on $\mathbb{C}$, where it underlies the hyperbolic-volume and scissors-congruence applications; formalizing D and its five-term relation is the complex-analytic sequel.

3. **Universality.** Wojtkowiak's theorem – that every rational one-variable functional equation reduces to the five-term – would be a deep and rewarding formalization target, of which the present result is the indispensable first brick.

Reproducing

```
git clone https://github.com/karlesmarin/godsil-gutman-lean
cd godsil-gutman-lean && lake exe cache get
lake build Dilog.FiveTerm
```

Lean toolchain `v4.30.0-rc2`, Mathlib pinned at `701fb6e9`. The five-term relation is `Dilog.rogersL_fiveterm` in `Dilog/FiveTerm.lean`; its supporting lemmas are `hasDerivAt_rogersL`, `fiveterm_hasDerivAt_zero`, `tendsto_rogersL_nhdsGT_zero` and `continuousAt_rogersL`. Each is checked by `#print axioms` to depend only on `propext`, `Classical.choice`, `Quot.sound`.

References

[1] Z. Wojtkowiak, *The basic structure of polylogarithmic functional equations*, in *Structural Properties of Polylogarithms* (L. Lewin, ed.), Math. Surveys Monogr. **37**, AMS (1991), 205–231.

[2] R. de Jeu, *Towards regulator formulae for the K-theory of curves and number fields*, Compositio Math. **124** (2000), 137–194.

[3] D. Zagier, *The dilogarithm function*, in *Frontiers in Number Theory, Physics, and Geometry II*, Springer (2007), 3–65.

[4] T. Nakanishi, *Dilogarithm identities for conformal field theories and cluster algebras*, in *Symmetries, Integrable Systems and Representations*, Springer Proc. Math. Stat. **40** (2013), 407–444.

[5] L. J. Rogers, *On function sum theorems connected with the series $\sum x^n/n^2$*, Proc. London Math. Soc. **4** (1907), 169–189.

[6] L. Lewin, *Polylogarithms and Associated Functions*, North-Holland (1981).

[7] The Mathlib Community, *The Lean Mathematical Library*, CPP 2020, 367–381.